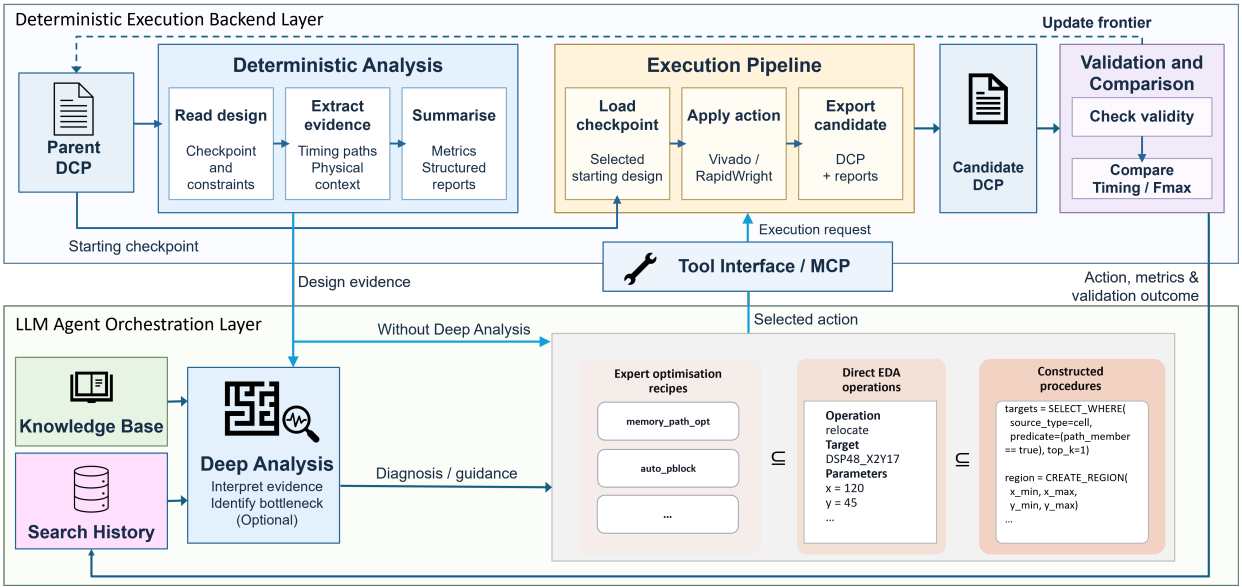

Deterministic Execution Backend Layer
Update frontier
Parent DCP
Deterministic Analysis
Read design
Checkpoint and constraints
Extract evidence
Timing paths
Physical context
Summarise
Metrics
Structured reports
Execution Pipeline
Load checkpoint
Selected starting design
Apply action
Vivado / RapidWright
Export candidate
DCP + reports
Candidate DCP
Validation and Comparison
Check validity
Compare Timing / Fmax
Starting checkpoint
Execution request
Tool Interface / MCP
Design evidence
Action, metrics & validation outcome
LLM Agent Orchestration Layer
Without Deep Analysis
Selected action
Knowledge Base
Search History
Deep Analysis
Interpret evidence
Identify bottleneck
(Optional)
Diagnosis / guidance
Expert optimisation recipes
memory_path_opt
auto_pblock
...
Direct EDA operations
Operation
relocate
Target
DSP48_X2Y17
Parameters
x = 120
y = 45
...
Constructed procedures
targets = SELECT_WHERE(
source_type=cell,
predicate=(path_member == true), top_k=1)
region = CREATE_REGION(
x_min, x_max,
y_min, y_max)
...